

Akkshay Sundara Rajan

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EDUCATION

Purdue University

Bachelor of Science, Computer Science and Mathematics (Double Major)

West Lafayette, IN

Aug 2024 — (Expected) Dec 2027

- Dean's List (4x), Semester Honors (4x), 3.92 GPA
- Coursework: Data Mining, Machine Learning, Data Structures and Algorithms, Linear Algebra, Statistics, DevOps Tools

EXPERIENCE

Applied AI Intern (Incoming)

Anunta

Jun 2026 — Aug 2026

Remote

Project Lead

Machine Learning @ Purdue - Student Organization

Sep 2025 — Present

West Lafayette, Indiana

- Led 6 developers in building an RL agent to play Pokemon competitively achieving 86% winrate against baseline heuristics.
- Designed a hierarchical BERT-style encoder stack from scratch including a custom game-state tokenizer, fused SwiGLU transformer layer with FlashAttention, recurrent memory tokens and a decoupled actor-critic path trained using PPO.
- Optimized rollout throughput to 40 actions/s using query coalescing, pinned memory and multi-threaded parallel envs.
- Dockerized and deployed the inference architecture to Google Compute Engine, handling 20+ concurrent battles per node.

Undergraduate Researcher / ML Engineer

Dr Philip Low's Lab

Aug 2025 — Dec 2025

West Lafayette, Indiana

- Built a distributed protein structure inference pipeline using NVIDIA Boltz-2, scaling and managing prediction workloads across 8 HPC nodes through SLURM scripts.
- Developed a postprocessing workflow (OpenMMDL) to test the stability of predicted complexes for drug discovery research.

Software Engineering Intern

Siemens Energy

Jun 2025 — Aug 2025

Gurgaon, India

- Developed an ETL pipeline using Python and SQL on Azure to manage and analyze 10 million sensor data records.
- Implemented an early-warning system using decision trees in scikit-learn to detect high outlet temperatures in gas turbines.
- Created a FFT-based diagnosis system to classify turbine failure modes by matching sensor data against past failures.
- Automated the visualization of multi sensor feeds in a dashboard using plotly, dash and Flask saving 1 hour per inspection.

Undergraduate Researcher

AIDA³ Labs, Purdue

Feb 2025 — Sep 2025

West Lafayette, Indiana

- Prototyped a medical emergency response agent using Llama 3 and LangChain to support dispatch coordination.
- Designed human-in-the-loop route planning workflows using Google Maps MCP data with LLM reasoning over call logs.

PROJECTS

sift - Multimodal Semantic Search Engine

github.com/akkshay0107/sift

- Engineered a high-performance semantic search engine for multimodal retrieval across text, image, audio, and video files.
- Built multimodal ingestion pipelines with OCR, audio transcription (Whisper), and Qwen3-VL to embed files into a local Qdrant vector database, optimized to minimize RAM consumption by 40% using BF16 quantization.
- Architected a real-time, event-driven indexing daemon using watchdog and content hashing for zero redundancy updates.

glimpse - AI Newsletter Automation Platform

github.com/sanjayriram44/glimpse

- Built a full-stack newsletter SaaS with FastAPI, Next.js, Supabase to consolidate newsletters into a daily audio digest.
- .Langchain, Gemini and ElevenLabs used to process emails into podcast style scripts with S3 for caching generated audio.

SKILLS

- **Programming Languages:** Python, C++, Rust, Java
- **ML/Data:** PyTorch, Langchain, Scikit-learn, HuggingFace, Pandas, Qdrant, Numpy, SQL, Postgres
- **Backend/Cloud:** FastAPI, Docker, Redis, Celery, Azure, AWS, Google Cloud Platform, SLURM, Bash